Alexander Shaffer

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Education

2025-Present **PhD in Computer Science**, *Scientific Computing*,
University of Illinois at Urbana Champaign, *GPA – 4.0 out of 4.0*.
13 credit hours completed

2021-2025 **Bachelor of Science in Engineering Physics**, *Computational Track, Minor in Mathematics, Minor in Computer Science*,
University of Illinois at Urbana Champaign, *GPA – 3.78 out of 4.0*.

2017-2021 **High School Education**, *Urbana High School*, Urbana, Illinois, *GPA – 5.658*.
High School Diploma completed May 2021

Research

Currently I am a PhD student working with Paul Fischer and Luke Olson at the University of Illinois at Urbana Champaign. My research relates to the use of numerical methods, including spectral element methods, for solving the incompressible Navier-Stokes equations and incompressible magnetohydrodynamics (MHD) equations on large scales.

Experience

2024-2025 **Undergraduate Research Assistant**, iSens Lab,
University of Illinois at Urbana Champaign.
Working with the *iSens* Group `http://isens.cs.illinois.edu/`
Developing machine learning methods to predict wireless signal propagation in real time and visualize it using an augmented reality app.

2024 **Instructor**, Academy High, Champaign Illinois ,
Course website: `https://redhorn144.github.io/AHCryptography/`.
Designed and taught a once weekly high school elective course on cryptography to 12 students during the fall 2024 school semester.

Summer 2024 **Undergraduate Research Assistant**, *Center for Exascale Enabled Scramjet Design (CEESD)*, University of Illinois at Urbana Champaign, Worked with CEESD during the summer of 2024 helping develop fluid dynamic simulations used for modeling scramjet engines. This involved numerically solving equations for fluid dynamics using discontinuous Galerkin methods and evaluating performance of those simulations.

Spring 2024 **Independent Study in Mathematics**, Explored the use of radial basis function (RBF) methods in Non-Euclidean spaces. Displayed the benefits of using a surface metric when forming RBF interpolants on surfaces. Implemented a variation on Kansa's (RBF collocation) method in order to solve PDE's on surfaces using geodesic distance as a surface metric., Code from this independent study may be found at `https://github.com/redhorn144/Non-Euclidean-RBF-and-Kansa`.

2021-2025 **Undergraduate Research Assistant**, Heliophysics Research Group, University of Illinois at Urbana Champaign.
Working with the *HeRA* Group `https://www.ilie.ece.illinois.edu/`
Currently producing data-driven mathematical models for analyzing and visualizing properties of ionospheric data in order to better understand space weather. Specifically, using Python, SciPy, and Plotly to develop interpolation schemes, evaluate their effectiveness, and visualize the results.

2022-2025 **Undergraduate VR Lab Developer**, Immersive Learning Laboratory, University of Illinois at Urbana Champaign.
Working with the Immersive Learning Laboratory to develop electricity and magnetism virtual reality (VR) experiences in Unity. These experiences are used for a lab course that is taken alongside the University of Illinois's Fields and Waves I course (ECE 329) `https://www.ilie.ece.illinois.edu/immersive-learning-lab`.

Awards

2025-Present **Paul Saylor Fellowship in Numerical Analysis & Scientific Computing**.

2023 **Best Magnetosphere-Ionosphere Coupling poster**, *Geospace Environment Modeling workshop*, Best out of 35 posters.

Skills

Programming python, MPI, C++, C#, Matlab

Miscellaneous Large scale fluid and Magnetohydrodynamics (MHD) simulations with Nek5000 and NekRS, Data visualization using Plotly & VTK, Git, Virtual Reality development in Unity

Software Products **Lazyfit Python Library**, *Creator and maintainer of the lazyfit python library which may be used to create and optimize radial basis function (RBF) interpolants for mesh-free data. This expands upon existing implementations of RBF interpolation, such as scipy's, by making use of efficient error estimation for interpolant optimization. Roughly 3100 downloads.*, Library: `https://pypi.org/project/lazyfit/`, Published December 2023.

Publications

2024 **Visualizing Polarization Effects of Gravitational Waves Using Particle Rings and Surfaces in Virtual Reality**, *K. Schumacher, S. Joshi, D. Srivastava, A. Shaffer, A. Jog, J. Shih, E. Shaffer.*, Advances in Visual Computing, 19th International Symposium, ISVC 2024, Lake Tahoe, NV, USA, October 21–23, 2024, Proceedings, Part I.

2025 **Building Robust VR Learning Environments: Best Methods to Visualize Divergence-Free Vector Fields**, *A. Shaffer, R. Ilie*, ASEE PEER, 2025 ASEE Annual Conference & Exposition, Montreal, Quebec, Canada, June 22-25, 2025.

Presentations

2025 **American Society for Engineering Education (ASEE) 2025 Conference**, *Building Robust VR Learning Environments: Best Methods to Visualize Divergence-Free Vector Fields*.

Presented this paper as a talk at ASEE 2025. This paper surveys methods for visualizing magnetic fields in virtual reality (VR). It implements an efficient real time methods for solving for magnetic field lines in a VR environment.

2024 **International Symposium in Visual Computing (ISVC)**, *Visualizing Polarization Effects of Gravitational Waves Using Particle Rings and Surfaces in Virtual Reality*.

Presented this paper as a talk at ISVC 2024. This paper covers a novel visualization of gravitational waves in virtual reality. The visualization incorporates polarizations from the most popular modified theories of gravity, as well as novel visualization metaphors.

2023 **Geospace Environment Modeling Workshop**, *A 3D Visual Reconstruction of Ionospheric Outflow*.

An expansion on the research presented at the 2022 Geospace Environment Modeling (GEM) workshop. Evaluation of interpolation error was greatly expanded, including a measure of altitude dependent interpolation error showing that accuracy increased with altitude in general. The ionosphere visualizations were also improved, incorporating improved colormaps, and improved visualizations styles. Gave a poster presentation at the GEM workshop about visualizing ionospheric plasma using radial basis function interpolation to generate plasma data at unsampled locations making the data suitable for visualization. **Received the award for the best Best Magnetosphere-Ionosphere Coupling poster**

2022 **Geospace Environment Modeling Workshop**, *A 3D Interactive Visualization Tool for Polar Wind Outflow Simulations*.

Gave a poster presentation at the Geospace Environment Modeling (GEM) workshop about visualizing ionospheric plasma by using radial basis function interpolation.

2022 **Center for Geospace Storms Workshop**, *Expanding Methods of Ionospheric Plasma Visualization*.

Gave a talk at the Center for Geospace Storms Workshop regarding visualization of ionospheric plasma by using radial basis function interpolation for visualization. Created plots displaying the regions in which nitrogen ions were dominant and in which oxygen ions were dominant, helping display the spatial dependence on methods of ionospheric outflow.

2022 **American Geophysical Union Fall Meeting**.

Exhibited at the American Geophysical Union Fall Meeting with the Immersive Learning Laboratory multiple virtual reality labs regarding Electricity and Magnetism. Two virtual reality experiences were demoed, one about Gauss's law and one about wave polarization.

2022 **Undergraduate Research Symposium**, *The Application of Radial Basis Function Interpolation to the Analysis of Polar Region Atmospheric Ion Density*, University of Illinois.

Presented a poster at the University of Illinois Undergraduate Research Symposium. The presentation was a comparison of different radial basis functions for the purpose of interpolation to recreate ionospheric plasma data.

Relevant Coursework

Mathematics, Math 241 Vector Calculus, Math 441 Differential Equations, Math 415 Linear Algebra, Math 423 Differential Geometry, Math 453 Number Theory, Math 447 Real Variables, Math 442 Intro to Partial Differential Equations.

Physics, Physics 325 Classical Mechanics I, Physics 225 Relativity and Math Applications, Physics 435 Advanced Electromagnetism I, Physics 436 Advanced Electromagnetism II, Physics 486 Quantum Physics I, Physics 486 Quantum Physics II, Physics 446 Computational Physics, Physics 402 Light.

Computer Science, Computer Science 555 Numerical Methods for PDEs, Computer Science 554 Parallel Numerical Algorithms, Computer Science 498 Intro to Quantum Computing, Computer Science 124 Intro to Computer Science I, Computer Science 128 Intro to Computer Science II, Computer Science 173 Discrete structures, Computer Science 225 Data Structures, Computer Science 519 Scientific Visualization, Computer Science 450 Numerical Analysis.

Electrical and Computer Engineering, ECE 598 EM of Continuum Media.

Theoretical and Applied Mechanics, TAM 570 Computational Fluid Mechanics.